IAT New Test 24

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

Instructions

Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

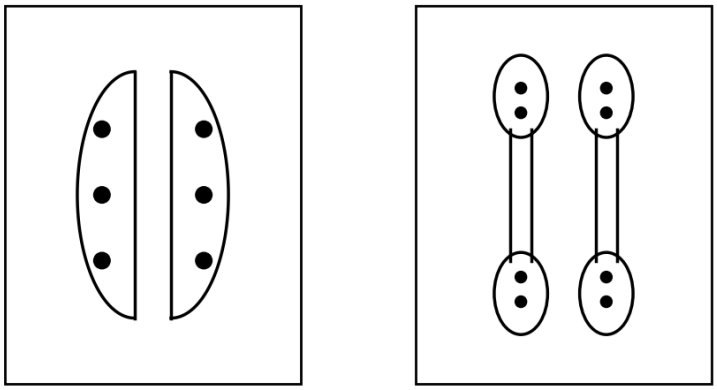
Q1 Three vertebrates are compared. P has a three-chambered heart, moist glandular skin and a larval stage with gills; Q has dry scales, a four-chambered heart and lays shelled eggs on land; R has feathers and a four-chambered heart. Which classification uses the combination of traits without treating heart-chamber number as sufficient?

A) P Reptilia; Q Aves; R Amphibia
B) P Amphibia; Q Aves; R Reptilia
C) P Amphibia; Q Reptilia, represented by a crocodilian; R Aves
D) P Pisces; Q Mammalia; R Aves

Q2 The diagram compares two mature epidermal stomatal complexes. P has kidney-shaped guard cells and Q has dumbbell-shaped guard cells; the surrounding epidermal cells lack the dark chloroplast bodies shown in the guard cells. Which interpretation is supported?

Epidermal stomatal complexes (schematic)

Dark bodies: chloroplasts in guard cells



P Q

A) Both regulate a pore, although guard-cell geometry differs; chloroplast absence in ordinary epidermal cells does not imply absence from guard cells
B) Only P is a stoma because all guard cells must be kidney-shaped
C) The dark bodies establish that every epidermal cell is photosynthetic
D) Q must be a xylem vessel because its guard cells are elongated

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Q3 Two identical enzyme preparations are assayed at the same optimal pH and temperature with saturating substrate. Preparation Q contains twice as much active enzyme as P. Initial rates are measured before appreciable substrate depletion. Which pair of changes is expected for Q relative to P?

A) Maximum rate unchanged; substrate concentration needed for half-maximum rate doubled
B) Maximum rate halved; half-maximum substrate concentration unchanged
C) Both maximum rate and half-maximum substrate concentration doubled
D) Maximum rate doubled; half-maximum substrate concentration unchanged

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Q4 A model plant cell lineage starts with one dividing cell. In lineage P each division produces one cell that keeps dividing and one cell that permanently differentiates. In lineage Q both daughters keep dividing. Cells divide once per equal interval, with no death. After four intervals, what are the total cell counts P and Q, and the corresponding growth patterns?
A) P=4, Q=8; both arithmetic
B) P=5, Q=16; arithmetic and geometric respectively
C) P=8, Q=16; geometric and arithmetic respectively
D) P=16, Q=5; geometric and arithmetic respectively

Q5 An isolated axon initially has normal ion gradients. A brief treatment blocks the sodium–potassium pump but does not block voltage-gated channels. Before the gradients substantially dissipate, a threshold stimulus is applied. Which statement best separates the action potential from long-term gradient maintenance?
A) An initial action potential can still occur through channel-mediated ion movement, although prolonged pump inhibition will impair maintenance of the gradients
B) The pump must carry the inward sodium current of every depolarising spike
C) Potassium must enter the axon through the pump to generate the rapid repolarising current
D) Blocking the pump immediately makes both sides of the membrane have identical ion concentrations

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Q6 A couple correctly uses oral contraceptive pills. They believe this also protects them from sexually transmitted infections. Which advice correctly distinguishes contraceptive action from infection prevention?
A) Hormonal suppression of ovulation necessarily prevents entry of HIV
B) An intrauterine device should replace the pill because it reliably blocks all sexually transmitted pathogens
C) Continue suitable contraception if desired, but use condoms to reduce sexual transmission risk; hormonal contraception alone does not provide that barrier
D) Condoms prevent infection only when ovulation is simultaneously suppressed

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Q7 Ignore exceptional sex-determination mechanisms. A hen produces Z-bearing or W-bearing eggs; a human female produces X-bearing eggs; a queen honeybee can lay fertilised or unfertilised eggs. Which comparison is correct?
A) The male determines chromosomal sex in all three systems
B) Only the female's egg type matters in both birds and humans
C) An unfertilised honeybee egg develops into a diploid female
D) The egg's Z/W contribution determines bird sex, the sperm's X/Y contribution determines human sex, and fertilisation normally distinguishes diploid female from haploid male honeybees

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Q8 Three patients are investigated. P develops a rapid wheal after a harmless pollen challenge; a matched test without pollen is negative and allergen-specific IgE is elevated. Q has antibodies reacting with the patient's own joint tissue and inflammatory injury. R repeatedly develops opportunistic infections and has severely reduced helper T-cell counts. Which assignment fits the targets and response evidence?
A) P autoimmunity; Q allergy; R enhanced innate defence
B) P allergy; Q autoimmunity; R immune deficiency
C) P immune deficiency; Q allergy; R autoimmunity
D) All three are explained by antibodies attacking pollen

Q9 Three DNA-delivery tasks are planned: P introduces DNA into chemically treated bacterial cells; Q delivers DNA directly into the nucleus of an animal cell; R delivers DNA across a plant cell wall without first making protoplasts. Which methods match these physical barriers?
A) P microinjection; Q biolistics; R heat-shock transformation
B) P biolistics; Q chemical competence; R microinjection into a bacterial nucleus
C) P competence treatment followed by heat shock; Q microinjection; R DNA-coated particle bombardment
D) P chemical competence; Q heat shock only; R removal of all plant walls by the gene gun

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Q10 A lake contains a few very large fish and thousands of tiny plankton, while a rock face carries a spreading lichen mat whose individuals cannot be separated reliably. Which measurements best answer respectively: fish contribution to living mass, plankton abundance, and extent of lichen occupation?
A) Fish biomass; plankton number per volume; lichen percentage cover
B) Fish number alone; plankton average length alone; lichen number of disputed individuals
C) Fish occupied surface area; plankton colour; lichen water content
D) Use individual count alone in all three cases, because population density can never use biomass or cover

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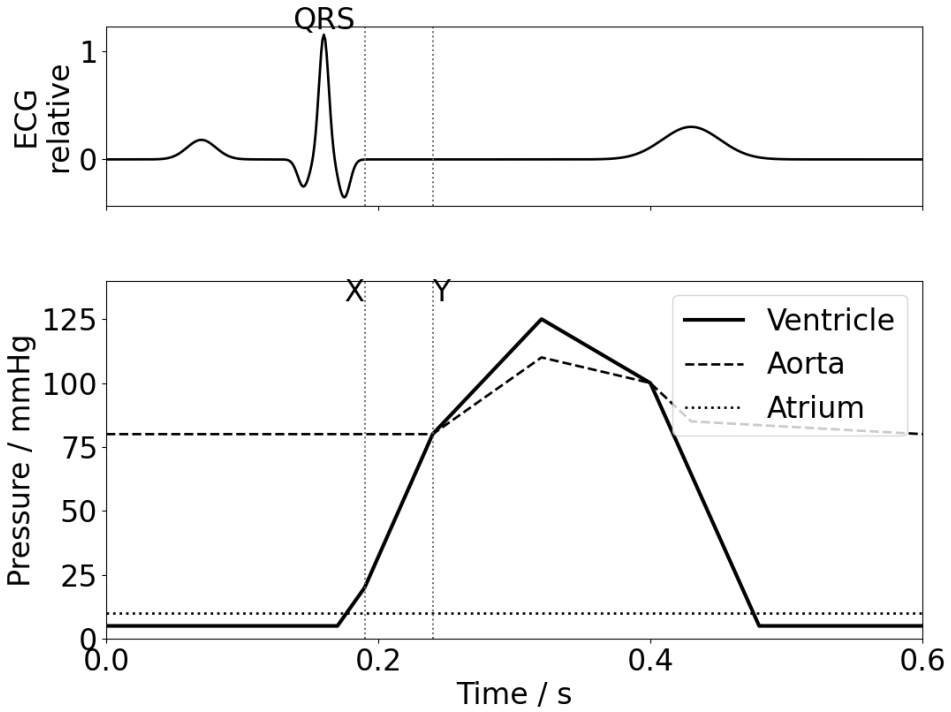
Q11 Two parents with blood group AB have a child. Assume the usual ABO alleles and no rare phenotypes. Which statement correctly links the possible child phenotypes to the meaning of multiple alleles and codominance?
A) Their child can be O because AB parents transmit the recessive allele i
B) Every child must be AB because the parental alleles blend permanently
C) Each parent contains all three alleles IA, IB and i in the same diploid locus
D) A, AB and B children are possible; IA and IB are codominant in an AB individual, while the population contains more than two alleles at the locus

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Q12 In the same species, two separate linked marker pairs P–Q and R–S give recombinant fractions of 0.10 and 0.45 in large test crosses, with no viability differences. Which inference is justified without additional physical or multi-marker data?
A) R–S must occupy different chromosomes because their recombinant fraction exceeds 0.25
B) P–Q show stronger observed linkage; the 0.45 value does not establish that the physical distance is exactly 4.5 times as large
C) P–Q must be physically farther apart because fewer recombinants survived
D) The physical distance of either pair is fixed uniquely by its recombinant fraction even if multiple crossovers occur

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Q13 The aligned trace shows a normal ECG and left-ventricular and aortic pressures. At marker X the QRS complex has just occurred, ventricular pressure is rising above left-atrial pressure but remains below aortic pressure. At Y the ventricular pressure first exceeds aortic pressure. Which sequence is consistent with excitation and valve mechanics?



A) X marks ventricular repolarisation and immediate semilunar opening; Y marks atrial depolarisation

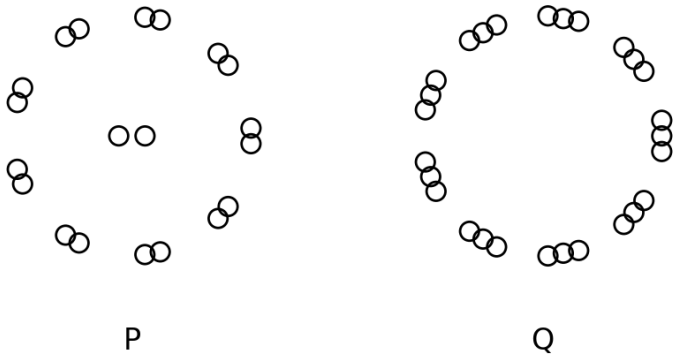
B) The QRS complex is blood ejection itself, so the aortic valve must already be open at X

C) Ventricular depolarisation precedes pressure development; X can be isovolumetric contraction with both left-heart valves shut, and ejection begins near Y

D) The mitral valve stays open throughout X to Y because ventricular pressure is increasing

Q14 Two transverse microtubule schematics are shown. P has nine peripheral doublets and a central pair; Q has nine peripheral triplets and no central pair. Which interpretation correctly relates these organisations?

Transverse microtubule organisations

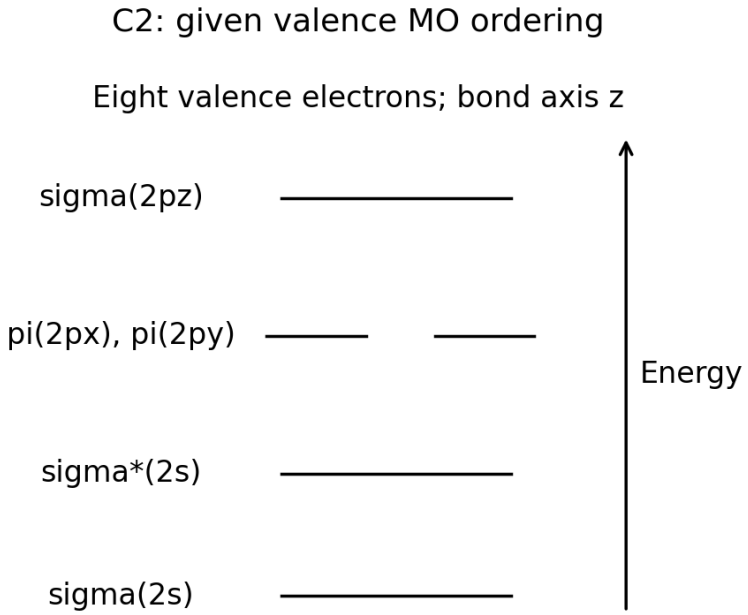


A) P is a typical eukaryotic ciliary or flagellar axoneme; Q is a centriole or basal-body pattern
B) P is a bacterial flagellum; Q is the 9+2 axoneme of a eukaryotic cilium
C) Both are identical because the number of peripheral groups is nine
D) P must contain 27 peripheral microtubules and Q only 18

Q15 A forest's dead leaves enter a detritus food chain, while some living leaves are eaten by grazing herbivores. Which statement correctly describes the relation between the two routes and the energy source?
A) The detritus route creates new solar energy, so it needs no earlier producers
B) Energy consumed by decomposers is completely recycled back into sunlight
C) Only living-leaf feeding can transfer energy to consumers
D) Both routes can transfer energy originally fixed by producers; matter can recycle, but energy is progressively dissipated rather than regenerated in a closed biological cycle

Chemistry

Q16 For ground-state C2, use the valence molecular-orbital order shown: sigma(2s), sigma*(2s), pi(2px)=pi(2py), sigma(2pz). The bond axis is z. Fill eight valence electrons according to the usual rules. Which statement follows within this MO model?



A) Bond order 2, diamagnetic, with the occupied 2p bonding contribution entirely from the two pi orbitals
B) Bond order 2, paramagnetic, with two unpaired 2p electrons
C) Bond order 1, diamagnetic, because every filled 2p orbital is antibonding
D) Bond order 3, diamagnetic, with a filled sigma(2pz) orbital

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Q17 Let Y4− denote fully deprotonated EDTA with two amine nitrogen donors and four carboxylate oxygen donors binding one Ca2+ ion. A stable colourless complex forms. Which description is correct?
A) Y4− is bidentate and the complex has charge +2
B) Y4− is hexadentate and the complex has charge −2; absence of visible colour does not show absence of coordination
C) Y4− is tetradentate because only charged atoms can donate a lone pair
D) The complex must be neutral because calcium is not a transition metal

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Q18 On heating, potassium permanganate gives potassium manganate, manganese dioxide and oxygen. In the balanced equation containing two KMnO4 units, what is the coefficient of O2 and how many electrons are transferred internally from oxygen to manganese?
A) O2 coefficient 2; 8 electrons
B) O2 coefficient 1/2; 2 electrons
C) O2 coefficient 1; 4 electrons
D) O2 coefficient 1; no electron transfer because all products are neutral compounds

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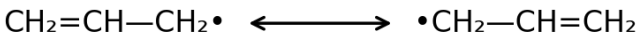
Q19 Three Period-3 oxides are tested. P reacts with acid but not aqueous alkali; Q reacts with both; R reacts with alkali but not acid under the stated test conditions. The candidate parent configurations are [Ne]3s1, [Ne]3s2 3p1 and [Ne]3s2 3p2. Which assignment best fits the usual acid–base trend?
A) P silicon oxide; Q sodium oxide; R aluminium oxide
B) P aluminium oxide; Q silicon oxide; R sodium oxide
C) P sodium oxide; Q silicon oxide; R aluminium oxide
D) P sodium oxide; Q aluminium oxide; R silicon oxide

Q20 For a one-dimensional uncertainty comparison, an electron has position uncertainty a and a proton has position uncertainty a/1000. Take mp=1836me and use the Heisenberg lower bound at equality for both. Define the associated kinetic-energy scale as (delta p)^2/(2m). What are the proton/electron ratios of velocity uncertainty and this energy scale?
A) 1000; 1000
B) 1000/1836; 1000000/1836
C) 1836/1000; 1836/1000000
D) 1/1836; 1/1836

Q21 A 0.500 g impure NaCl sample contains only a chloride-free inert impurity. Its chloride is completely precipitated as AgCl. The collected solid weighs 1.248 g before drying and 1.148 g after drying to constant mass; the loss is only water. Take M(AgCl)=143.5 and M(NaCl)=58.5 g mol−1. What NaCl mass percentage follows, and what error would using the wet mass cause?
A) 93.6%; it would overestimate purity
B) 93.6%; it would underestimate purity
C) 100%; drying cannot change a gravimetric inference
D) 80.0%; it would overestimate purity

Q22 The diagram shows two valid resonance contributors of an allylic carbon radical, with the double bond and single unpaired electron on opposite terminal positions. What remains unchanged between the contributors?

Allylic radical: two contributors

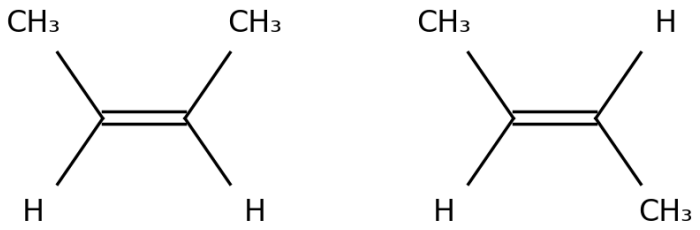


Same three-carbon framework

A) The location of every pi electron is fixed, so the drawings must be different conformers
B) Atom connectivity, total charge and the number of unpaired electrons remain unchanged; the actual radical is delocalised
C) One contributor must be a cation and the other an anion
D) The radical repeatedly converts into two separately isolable compounds at equilibrium

Q23 The figure shows cis- and trans-but-2-ene separately. Under conditions that give stereospecific anti addition of Br2, with no rearrangement or subsequent epimerisation, which product comparison is correct? Count configurational stereoisomers, including both members of a racemate.

Two but-2-ene geometries



cis trans

A) Cis gives only the meso form; trans gives an enantiomeric pair
B) Both give the same single optically active stereoisomer
C) Both give four stereoisomers because two stereocentres always give four distinct products
D) Cis gives an enantiomeric pair; trans gives the meso form of 2,3-dibromobutane

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Q24 A chloroform sample is exposed to air and light, producing a toxic oxidation product, carbonyl chloride. A separate sample is stored in a dark, tightly stoppered bottle containing a little ethanol. Which explanation correctly assigns the roles of the precautions?
A) Dark storage increases oxidation; ethanol simply replaces chlorine atoms by hydrogen
B) Stoppering alone removes every oxygen molecule already dissolved, so the other precautions have no chemical role
C) Darkness and restricted air access limit phosgene formation; ethanol can react with phosgene that forms, rather than making oxidised chloroform safe by dilution alone
D) Ethanol prevents oxidation because it converts all chloroform into an ionic chloride salt

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Q25 Phenol is subjected to two independent preparations. Route P uses CHCl3/aqueous NaOH followed by acid work-up; route Q uses sodium phenoxide with CO2 under pressure followed by acid work-up. Considering the usual major ortho products, which pair of product groups and separate tests is correct?
A) P has an aldehyde and gives Tollens positive but bicarbonate negative; Q has a carboxyl group and gives bicarbonate positive but ordinary Tollens negative
B) P and Q both give salicylic acid because both add one carbon
C) P gives an ether and Q an aldehyde; neither can release CO2 from bicarbonate
D) P gives an aldehyde but cannot reduce Tollens reagent because the ring is aromatic; Q is neutral to bicarbonate

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Q26 A peptide hormone and a steroid hormone are assayed separately before and after exhaustive protease treatment. The peptide hormone loses activity; the steroid hormone retains activity under the same treatment. Which conclusion is best supported?
A) Every hormone is a protein, and the steroid result disproves enzyme specificity
B) Different hormone classes can have different chemical structures; peptide-bond hydrolysis can disrupt the peptide hormone without cleaving a steroid into amino acids
C) The steroid hormone must be an inorganic salt because protease did not inactivate it
D) Hormones function principally as bulk energy stores, so protease effects cannot concern signalling

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Q27 For carbonic acid, take Ka1=5×10−7, Ka2=5×10−11 and Kw=10−14. In an aqueous NaHCO3 solution where water's own ionisation is not dominant, which comparison correctly explains the usual pH above 7?
A) HCO3− cannot donate a proton because it is negatively charged
B) Every hydrogen-containing salt must give an acidic solution
C) HCO3− is amphiprotic, but its basic constant Kw/Ka1=2×10−8 exceeds its acidic constant Ka2
D) Sodium ions hydrolyse to produce all the hydroxide

Q28 A dilute nonreacting gas follows p=KH x. At T1 its Henry constant is K and its mole fraction over a gas mixture is x1. At T2 the constant becomes 4K, total pressure doubles, and the gas-phase mole fraction of this gas changes from 0.20 to 0.40. What is x2/x1, and does total pressure alone predict the solubility change?
A) 1; no, because both partial pressure and KH change
B) 1/2; yes, total pressure alone is sufficient
C) 2; no, because KH is constant
D) 4; yes, because gas-phase composition cannot matter

Q29 Two electrolytic cells operate in parallel for 9650 s. The copper cell carries a constant 0.20 A and deposits Cu from Cu2+; the silver cell carries 0.30 A and deposits Ag from Ag+. Current efficiencies are 100%. Use F=96500 C mol−1 and atomic masses Cu=63.5, Ag=108. What are the deposited masses Cu and Ag?
A) 1.270 g and 3.240 g
B) 0.635 g and 1.620 g
C) 0.79375 g and 2.700 g
D) 0.635 g and 3.240 g

Q30 At a fixed temperature and composition, reaction R has delta G=+20 kJ per written reaction and S has delta G=−50 kJ per written reaction. An ideal coupling mechanism carries out exactly two R events per S event. What is delta G per coupled event, and what does it imply?
A) −30 kJ; both isolated reactions must be spontaneous
B) +30 kJ; coupling can never alter thermodynamic feasibility
C) −10 kJ; R becomes individually spontaneous even when uncoupled
D) −10 kJ; the specified combined process is thermodynamically favourable although uncoupled R is not

Mathematics

Q31 Two circles centred at the origin have radii 5 and sqrt(13). A line through (0,5) cuts both circles into chords whose lengths differ by 4. What is the product of the slopes of all such lines?
A) −9/16
B) 16/9
C) −16/9
D) −4/3

Q32 A right-opening parabola has horizontal axis and directrix x=−1. The line y=x is tangent at (2,2). What is the focus?
A) (1/2,−1)
B) (2,2)
C) (−1,−1)
D) (2,−1)

Q33 For f(x)=x^x on x>0, which statement correctly describes its stationary point and curvature?
A) It has one stationary point x=e^(−1), which is a global minimum; f” is positive throughout its domain
B) It has a local maximum at x=e^(−1), with f” negative there
C) It has stationary points at x=0 and x=1
D) It is increasing everywhere because its base and value are positive

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Q34 For a≥0 define F(a)=the integral from 0 to 1 of 1/(1+ax) dx. How many values of a satisfy F(a)=ln 2?
A) Zero
B) Two
C) Infinitely many
D) Exactly one, namely a=1

Q35 Let f(x)=(x−1)/x. Consider real starting values for which every successive iterate is defined. Which statement about the smallest positive iterate returning to the starting value is correct?
A) The allowed set is R excluding 0 and 1, and every allowed point has period 3
B) Every real point has period 2
C) The allowed set excludes only 0, and every allowed point is fixed
D) Only x=1 has period 3

Q36 Let A have columns c1,c2,c3 and rows (1,1,0),(0,1,1),(1,0,1). Form B with columns c1,c2+2c1,c3. What is the first column of B inverse?
A) (1/2,−1/2,1/2) transpose
B) (−1/2,1/2,−1/2) transpose
C) (1/2,1/2,−1/2) transpose
D) (−1,1,−1) transpose

Q37 Three distinguishable fair six-sided dice are thrown. Given that the minimum of the three results is exactly 2, what is the probability that their sum is 8?
A) 1/10
B) 6/125
C) 3/61
D) 6/61

Q38 Expand (1−2x)^8 and then set x=1/4. Label terms by r=0,...,8, where term r contains x^r. What are the indices of the two greatest-magnitude terms, and their signed sum after substitution?
A) r=3,4; sum 14
B) r=2,3; sum 0
C) r=2,3; sum 14
D) r=1,2; sum 3

Q39 The arithmetic progression is 3,5,7,... . For a positive integer n, its alternating sum a1−a2+a3−...+(−1)^(n−1)an is 11. What is an?
A) 11
B) 17
C) 19
D) 21

Q40 For each fixed real a define L(a)=lim as x tends to 0 of [sqrt(1+ax)−1]/[x(x−a)], taking x through points where the quotient is defined. What are L(0) and lim as a tends to 0 of L(a), respectively?
A) 0 and −1/2
B) −1/2 and −1/2
C) 0 and 0
D) 1/2 and 0

Q41 The planes are x+y+z=1, x−y+z=0 and 2x+az=b. For which a,b is their common intersection a line, and what is that line's distance from the origin?
A) a=2,b arbitrary; distance 1/2
B) a=1,b=2; distance sqrt(6)/4
C) a=2,b=1; distance sqrt(6)/4
D) a=2,b=1; distance sqrt(2)/2

Q42 Six distinct beads A–F form a regular hexagonal necklace. A and B must occupy opposite vertices. Arrangements related by a rotation or a reflection of the whole necklace are identified. How many arrangements are there?
A) 24
B) 12
C) 6
D) 48

Q43 For z nonzero, the complex number z+1/z is real. What is the complete locus of z?
A) The nonzero real axis together with the unit circle
B) The unit circle only
C) The imaginary axis excluding zero
D) The real axis including zero and no other points

Q44 Consider nonnegative continuously differentiable functions y on x≥0 satisfying $(y')^2=4y$ and y(0)=0, with a right derivative at zero. Which statement is correct?
A) Only $y=x^2$ satisfies the initial-value problem
B) Infinitely many solutions exist; for every c≥0, y=0 on 0≤x≤c and $y=(x-c)^2$ on x≥c is one
C) Only y=0 satisfies the equation because the initial derivative is zero
D) No solution exists because dividing by sqrt(y) is impossible at x=0

Q45 For 0≤theta≤2pi, when is arcsin(sin theta)=arccos(cos theta), using the principal ranges [−pi/2,pi/2] and [0,pi]?
A) For every theta in [0,pi]
B) Only theta=pi/2
C) For theta in [0,pi/2] together with theta=2pi
D) For theta in [pi,3pi/2]

Physics

Q46 A particle of mass m is stopped from speed v0 by either a constant opposing force F or an opposing agency removing kinetic energy at constant power P. Ignore other forces and use the ideal models until rest. If v0 doubles, by what factors do the two stopping distances increase, respectively?
A) 2 and 2
B) 4 and 4
C) 8 and 4
D) 4 and 8

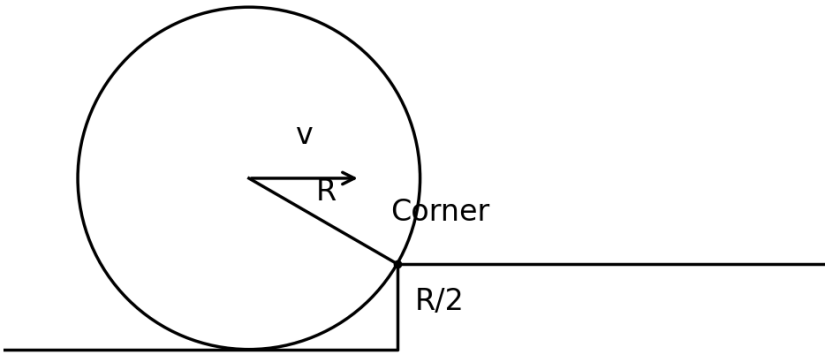
Q47 A proposed power law is P=alpha v^2 exp(−beta x)+gamma sqrt(a x), where x is a positive displacement, v speed and a positive acceleration. Which dimensions are required for alpha, beta and gamma, and does dimensional consistency alone establish that the law is physically correct? Use M,L,T for mass, length and time.
A) alpha: M T−1; beta: L; gamma: M L T−2; yes, dimensions prove the law
B) alpha: M L^2 T−3; beta: L−1; gamma: M T−1; no
C) alpha: M T−1; beta: L−1; gamma: M L T−2; no, consistency is necessary but not sufficient
D) alpha: M T−2; beta: L−1; gamma: M L T−1; yes, any consistent expression is a verified law

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Q48 Eight identical spherical liquid drops of radius r coalesce into one sphere without evaporation. Surface tension and the density difference from the surrounding viscous fluid stay constant. All drops are in the Stokes regime before and after coalescence. What are the final/initial ratios for terminal speed of one drop and total surface energy of the eight-drop system?
A) 4 and 1/2
B) 2 and 1/4
C) 4 and 2
D) 8 and 1/2

Q49 A solid wheel of mass m, radius R and central moment $mR^2/2$ rolls rightward at speed v into a step of height R/2. In the brief impact it sticks without slipping to the corner shown and loses contact with the ground. Neglect gravity's impulse. It then rotates about that fixed corner without further dissipation. What minimum incoming speed just lets its centre rise above the corner to the top of the step?

Wheel approaching a step

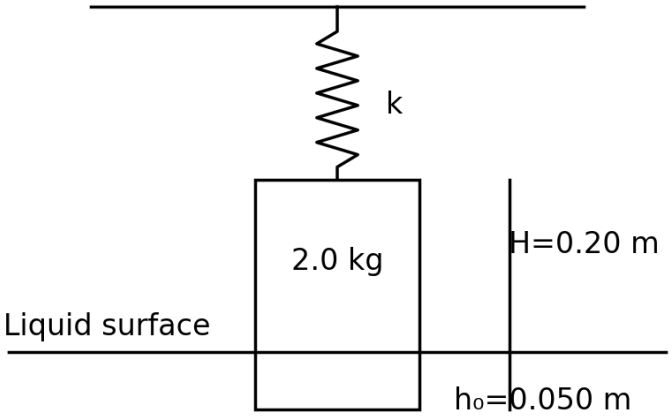


A) sqrt(gR)
B) sqrt(3gR/2)
C) sqrt(2gR/3)
D) sqrt(2gR)

Q50 One mole of monatomic ideal gas follows a quasistatic cycle: isochoric heating from (T,V) to (2T,V), isothermal expansion to (2T,2V), isochoric cooling to (T,2V), then isothermal compression to (T,V). There is no regenerator; count all positive external heat as input. What is the efficiency?
A) ln2/(3/2+2 ln2)
B) 1/2
C) ln2/(3+ln2)
D) 1

Q51 A vertical spring of constant 100 N m−1 supports a 2.0 kg uniform block of horizontal cross-section 0.010 m² and height 0.20 m, partly immersed in a liquid of density 1000 kg m−3. At equilibrium its submerged depth is 0.050 m. Neglect viscosity and surface tension; take g=10 m s−2. Which statement about small vertical motion is correct?

Spring-supported block at equilibrium



A) Angular frequency is sqrt(50) rad s−1 and 0.060 m amplitude remains described by the same SHM law
B) Angular frequency is zero because spring and buoyancy balance weight
C) Angular frequency is 10 rad s−1; amplitude 0.060 m cannot remain governed by the same partly immersed linear restoring law throughout the cycle
D) Angular frequency is sqrt(50) rad s−1; buoyancy has no effect on restoring stiffness

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Q52 A tube closed by water at one end has its first two resonances with a 340 Hz source at air-column lengths 0.20 m and 0.70 m, with no resonance between them. Assume one constant open-end correction e. What are sound speed and e?
A) 170 m s−1; 0.050 m
B) 340 m s−1; 0.050 m
C) 340 m s−1; zero
D) 680 m s−1; 0.30 m

Q53 At a far-field point on an ideal electric dipole’s equatorial plane, excluding infinity, which combination is correct? Take potential zero at infinity and dipole moment p nonzero.
A) Potential and electric field both vanish
B) Potential is nonzero but field vanishes
C) Potential vanishes, but the field is nonzero and opposite to p
D) Both potential and field point along p

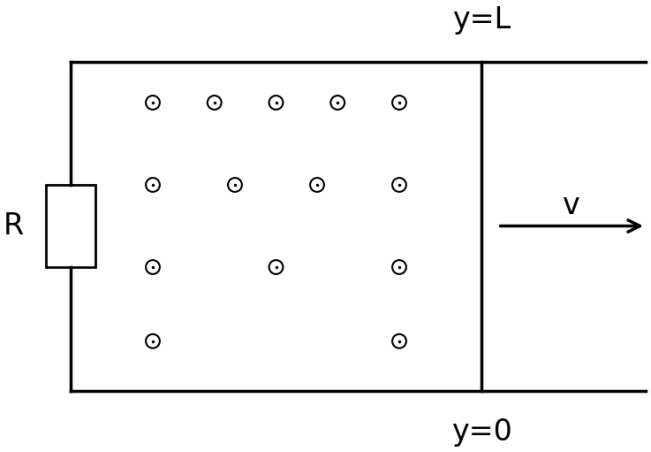
Q54 A resistor has R=10[1+0.010 delta T] ohm, with delta T in kelvin above ambient. Heat loss is 0.10 delta T watts. Compare an ideal 12 V supply with an ideal 1.20 A supply, each held constant. Within this linear thermal model, which steady-state conclusion is correct?
A) Both give delta T=80 K because their initial powers match
B) Constant voltage gives delta T=80 K; constant current has no nonnegative finite steady solution
C) Constant current gives delta T=80 K; constant voltage has no steady solution
D) Both remain at ambient because resistance increases

Q55 A sealed sample contains radioactive parent P and its stable daughter D only. Each parent decay makes one daughter, with no losses. Initially ND/NP=2; after time t the ratio is 11. What fraction of the initial parents remains, and how many parent half-lives have elapsed?
A) 1/11; log2 11
B) 1/12; log2 12
C) 2/11; log2(11/2)
D) 1/4; two

Q56 A conducting rod from y=0 to L moves at speed v in +x along a closed rail circuit of total resistance R. The static field is Bz(y)=B0(y/L)^2 in +z. Use B0=0.60 T, L=0.50 m, v=3.0 m s−1 and R=0.30 ohm. Neglect inductance. What are the rod's emf magnitude, positive end and external force needed for steady motion?

Moving rod in a transverse field gradient

$B_z = B_0(y/L)^2$, out of page



A) 0.90 V; y=L positive; 0.90 N in +x
B) 0.30 V; y=L positive; 0.10 N in −x
C) 0.90 V; y=0 positive; 0.30 N in +x
D) 0.30 V; y=0 positive; 0.10 N in +x

Q57 A thin symmetric biconvex glass lens has refractive index 1.50 and surface radii +20 cm and −20 cm. What are its focal lengths when immersed separately in liquids of indices 4/3 and 1.60? Use the relative-index lens-maker formula and unchanged curvatures.
A) +20 cm and +20 cm
B) +80 cm and −160 cm
C) −80 cm and +160 cm
D) +40 cm and −80 cm

Q58 An ideal resistor and capacitor are connected in parallel to a 10 V rms sinusoidal supply. The resistor current is 1.0 A rms, and the capacitor current is 1.0 A rms. What are the rms supply current and mean power drawn?
A) 2.0 A; 20 W
B) 1.0 A; 10 W
C) 2.0 A; zero
D) sqrt(2) A; 10 W

Q59 Samples P,Q,R are compared. P has no band gap and its resistivity rises with temperature. Q has a small gap, its carrier concentration rises strongly on heating, and electron and hole concentrations are equal; electron mobility exceeds hole mobility. R has a much larger gap and negligible carriers in the tested range. Which statement is correct?
A) P is metallic, Q is intrinsic semiconducting and R is insulating; Q can have unequal electron and hole current contributions despite equal carrier concentrations
B) Q is extrinsic n-type solely because its electrons move more readily
C) Equal electron and hole concentrations force equal mobilities and equal current contributions
D) P must be an insulator because its resistivity increases with temperature

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Q60 A slit of width a is imaged as a Fraunhofer diffraction pattern in a lens's back focal plane. In air, light of vacuum wavelength lambda0 and focal length f gives central width w. The whole optical region is then filled with index 1.50 liquid; the measured focal length becomes 1.50f and the slit width is unchanged. In the paraxial limit, what happens to central width?
A) It becomes 1.50w
B) It becomes w/1.50
C) It remains w because the wavelength decrease cancels the focal-length increase
D) It becomes 2.25w

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / C A crocodilian exception prevents four heart chambers from being used as a unique class marker

Answer: (C) P is an amphibian, Q a crocodilian reptile and R a bird.

Why: P combines the moist glandular covering and aquatic larval respiration characteristic of amphibians. Most reptiles have an incompletely divided ventricle, but crocodilians have a four-chambered heart; dry scales and shelled terrestrial eggs place Q with reptiles despite this exception. Feathers identify R as a bird. Thus the same heart-chamber count can occur in different vertebrate classes, and several independent characters must be combined.

Why not the others: (A) exchanges the amphibian and reptile coverings and ignores R's feathers. (B) assigns a featherless scaled animal to birds and a feathered animal to reptiles. (D) gills in an amphibian larva do not make the adult a fish, and Q lacks the diagnostic mammalian combination.

Key point: Classify using a coherent character set while retaining genuine exceptions to broad generalisations.

Q2 / A Guard-cell specialisation and alternative stomatal geometry distinguished from ordinary epidermal cells

Answer: (A) Different guard-cell shapes can operate the same stomatal function.

Why: Each complex consists of a pore bounded by two specialised guard cells. Kidney-shaped guard cells occur widely, whereas the dumbbell form is characteristic of grasses. Guard cells can contain chloroplasts even when the ordinary epidermal cells shown lack them. The diagram therefore compares specialisation within the epidermis rather than two different vascular tissues. It does not establish a photosynthetic function for every surrounding cell.

Why not the others: (B) incorrectly makes one common guard-cell shape universal. (C) transfers a feature of the specialised guard cells to all epidermal cells despite the observed contrast. (D) confuses a paired surface pore apparatus with an internal conducting vessel.

Key point: Identify a tissue component by its organisation and role, allowing specialised cells to differ from their neighbours.

Q3 / D Saturating initial rates scale with active-site number without changing the half-maximum substrate concentration

Answer: (D) Doubling active enzyme doubles maximum rate without changing the substrate dependence of each active site.

Why: With substrate saturating, the number of functioning catalytic sites limits the initial total rate. Q has twice as many equivalent sites, so it can process twice as much substrate per unit time. The molecular affinity and turnover properties of each enzyme molecule are unchanged under the matched conditions. Therefore the substrate concentration giving half of that preparation's own maximum remains the same. The initial-rate condition prevents later substrate depletion from obscuring this comparison.

Why not the others: (A) keeps the site-limited maximum fixed and wrongly changes the enzyme's substrate dependence. (B) reverses the effect of adding active catalytic sites. (C) confuses a vertical rate scaling with a change in the horizontal substrate-concentration scale.

Key point: Separate enzyme abundance from the properties of an individual catalytic site.

Q4 / B Division competence determines arithmetic versus geometric cell-lineage growth

Answer: (B) P contains 5 cells and Q 16 cells; their increments follow arithmetic and geometric growth respectively.

Given: Each lineage begins with one cell and undergoes four equal division intervals without loss.

Formula/Principle: Count only daughters retaining division competence when determining the next interval's increase.

Step 1: P always has one dividing cell. Each interval adds one differentiated cell, so

$$N_P = 1 + 4 = 5.$$

Step 2: Every Q cell divides, giving

$$N_Q = 1 \times 2^4 = 16.$$

Why: P's constant dividing compartment produces a constant absolute increase, whereas Q's expanding dividing compartment produces a constant proportional increase. Differentiated daughters remain in the total count but do not generate further descendants in this model.

Why not the others: (A) omits the initial cell and undercounts Q's doubling. (C) gives P geometric proliferation contrary to permanent differentiation. (D) exchanges the two lineage rules.

Key point: Growth depends on which descendants remain able to divide, not just the occurrence of mitosis.

Q5 / A An early spike can use existing ion gradients despite inhibition of their long-term pump maintenance

Answer: (A) Existing gradients can support an initial channel-mediated spike despite loss of their active maintenance.

Why: Rapid depolarisation normally involves increased sodium permeability through gated channels, followed by increased potassium permeability during repolarisation. These passive movements use electrochemical gradients already present before the stimulus. The sodium–potassium pump supports those gradients over longer periods but is not the gated pathway carrying each rapid spike current. The stipulated short observation interval matters: after extended inhibition, repeated activity and ion leakage would undermine the gradients and excitability.

Why not the others: (B) replaces the voltage-gated sodium pathway with the slower maintenance pump. (C) reverses the usual outward potassium movement during repolarisation and assigns it to the wrong mechanism. (D) assumes inhibition instantly removes previously established concentration differences.

Key point: Distinguish the pathway producing an electrical event from the process maintaining its energy source.

Q6 / C Preventing ovulation does not supply a barrier against sexually transmitted infection

Answer: (C) Hormonal contraception and a physical barrier address different routes to different outcomes.

Why: Oral contraceptives alter reproductive hormonal control and can prevent ovulation or otherwise reduce pregnancy probability. That action does not prevent contact with infectious secretions or pathogens during sexual intercourse. Correct condom use provides a barrier that reduces transmission risk as well as contributing to contraception. Risk reduction should not be described as an absolute guarantee, and neither pill use nor an intrauterine device substitutes for the relevant barrier.

Why not the others: (A) confuses blocking egg release with blocking pathogen entry. (B) incorrectly attributes universal infection protection to an intrauterine device. (D) makes a barrier's infection-prevention role depend on the ovarian cycle, although pathogen exposure does not require ovulation.

Key point: Evaluate pregnancy prevention and infection-risk reduction as separate biological mechanisms.

Q7 / D Bird ZW, human XY and honeybee haplodiploidy differ in the variable parental contribution

Answer: (D) The variable parental contribution differs among the three systems.

Why: Female birds are ZW and males ZZ, so alternative eggs provide the variable sex chromosome. In humans the female is XX and the male XY, making sperm type the variable chromosomal contribution. In honeybees, ordinary unfertilised eggs develop parthenogenetically into haploid males, whereas fertilised eggs develop into diploid females. These mechanisms cannot be collapsed into one universal rule about which parent determines sex.

Why not the others: (A) wrongly applies the human XY pattern to birds and haplodiploidy. (B) ignores the human male's alternative X- and Y-bearing sperm. (C) reverses both the ploidy and the usual sex of an unfertilised honeybee egg.

Key point: Identify the heterogametic parent or fertilisation rule before assigning a parental sex-determining role.

Q8 / B Controlled allergen evidence, self-antigen injury and helper-cell deficiency distinguish three immune failures

Answer: (B) P shows allergy, Q autoimmunity and R immune deficiency.

Why: P has an antigen-dependent excessive response to an ordinarily harmless environmental substance, consistent with an IgE-associated allergy. Q's response is directed against self tissue, which identifies loss of self tolerance rather than an exaggerated response to an external allergen. R has impaired helper T-cell availability together with infections that normally controlled immunity would resist. The matched negative challenge in P separates the observed pollen response from an unrelated skin reaction; the three cases therefore require both the target and the effectiveness of defence to be considered.

Why not the others: (A) exchanges the external and self targets and misreads reduced defence as enhancement. (C) reverses the evidence for P and R and assigns Q the wrong target class. (D) ignores the distinct self-directed response and T-cell deficit.

Key point: Distinguish excessive external-antigen responses, self-directed responses and insufficient protective responses.

Q9 / C Host barriers determine competence treatment, microinjection or biolistic DNA delivery

Answer: (C) The methods act on different barriers and target compartments.

Why: Chemical competence treatment can increase bacterial DNA uptake, with the appropriate heat-shock step aiding transformation. A fine micropipette can place DNA directly into an animal-cell nucleus during microinjection. Biolistics uses DNA-coated particles to cross plant-cell barriers, including walls, and does not require every target cell to be converted into a protoplast first. Successful delivery still does not by itself prove stable integration or expression.

Why not the others: (A) swaps direct animal-cell injection with plant particle delivery and gives the intact plant the bacterial procedure. (B) invokes a bacterial nucleus, which bacteria lack, and mismatches the targets. (D) wrongly treats heat shock as precise nuclear injection and claims a gene gun removes all walls.

Key point: Choose a delivery method from the cell barrier and intended destination before judging genetic success.

Q10 / A Biomass, numerical density and percentage cover answer different population questions

Answer: (A) Density can be expressed using the measure relevant to the organism and ecological question.

Why: Individual number can greatly understate the mass contribution of a few large organisms, so fish biomass addresses the first question directly. A standard-volume count gives a reproducible abundance measure for dispersed plankton. Percentage cover measures occupation by a lichen mat when boundaries between individuals are not operationally reliable. These measures are not interchangeable numerical units, but each can represent population presence or density for its specified purpose.

Why not the others: (B) supplies measurements that fail the mass, abundance or boundary constraints. (C) substitutes unrelated attributes for the requested quantities. (D) makes individual count universal despite the explicit difficulty of delimiting lichen individuals.

Key point: Define the biological quantity and sampling unit before choosing a density measure.

Q11 / D ABO offspring possibilities separate codominance within an individual from multiple alleles in a population

Answer: (D) The cross permits A, AB and B, and illustrates two distinct genetic terms.

Why: Each AB parent transmits either IA or IB. Their offspring can therefore be IAIA, IAIB or IBIB, producing A, AB or B phenotypes. IA and IB are both expressed in a heterozygous AB individual, which is codominance. Multiple allelism instead describes the existence of IA, IB and i in the population; one ordinary diploid individual still carries only two alleles at the locus. Neither parent in this cross carries i.

Why not the others: (A) invents an i allele absent from both parental genotypes. (B) treats codominance as permanent blending that abolishes segregation. (C) confuses the population's allele inventory with an individual's diploid genotype.

Key point: Separate allele transmission, heterozygous expression and population-level allele diversity.

Q12 / B Observed linkage strength does not uniquely determine a ratio of physical chromosome distances

Answer: (B) Recombinant fractions compare observed linkage but do not uniquely measure long physical distances.

Why: Fewer recombinant offspring indicate stronger observed association for P–Q under the stated equal-viability conditions. A fraction close to one-half can arise from widely separated loci on the same chromosome as well as independent assortment, so R–S need not lie on different chromosomes. Multiple exchanges can restore parental marker arrangements and go uncounted as recombinants. Therefore the two fractions cannot be converted into an exact ratio of DNA lengths using these data alone.

Why not the others: (A) invents a 0.25 boundary for different chromosomes. (C) contradicts both the linkage interpretation and the exclusion of differential survival. (D) ignores undetected multiple exchanges and variation in recombination along chromosomes.

Key point: Recombination is evidence for association, not an unlimited ruler for physical DNA distance.

Q13 / C ECG ventricular depolarisation precedes pressure-driven valve opening and actual ejection

Answer: (C) Electrical excitation precedes force and the pressure crossing that opens the outlet valve.

Why: QRS reflects ventricular depolarisation, not the volume of blood being ejected. As contraction develops, ventricular pressure closes the mitral valve once it exceeds atrial pressure. Until it also exceeds aortic pressure, the aortic valve remains closed, so the intervening contraction can occur with both valves shut and no ejection. The crossing near Y supplies the pressure condition for aortic opening and the start of ejection. This combines the ECG with the pressure evidence instead of treating them as identical events.

Why not the others: (A) confuses depolarisation with repolarisation and opens the outlet too early. (B) equates an electrical waveform with a mechanical flow event. (D) contradicts the reversed pressure difference across the mitral valve at X.

Key point: Relate electrical activation to subsequent mechanics through pressure differences, not by equating ECG peaks with blood flow.

Q14 / A Peripheral doublets with a central pair distinguished from a nine-triplet basal-body cylinder

Answer: (A) The doublet-plus-central-pair pattern differs from the triplet cylinder.

Why: The typical motile eukaryotic ciliary or flagellar axoneme has nine peripheral microtubule doublets and two central microtubules. Centrioles and basal bodies instead have nine peripheral triplets without that central pair. Nine peripheral groups alone therefore does not make the organisations identical. Bacterial flagella have a different protein organisation and are not built as the pictured 9+2 microtubule axoneme.

Why not the others: (B) exchanges the eukaryotic structures and incorrectly assigns microtubule architecture to a bacterial flagellum. (C) ignores doublet versus triplet membership and the centre. (D) reverses the peripheral totals: nine doublets contain 18 and nine triplets 27 microtubules.

Key point: Count both the peripheral group size and the central arrangement when identifying a microtubular structure.

Q15 / D Grazing and detritus pathways share producer-derived energy without cyclic energy regeneration

Answer: (D) Grazing and detritus routes share producer-fixed energy without creating a cyclic energy supply.

Why: A living leaf and its later litter both contain chemical energy derived from primary production. Herbivores can receive some through grazing, whereas decomposers and detritivores can receive some after material enters the detritus pathway. Transfers involve respiration and heat dissipation, so the energy flow requires continuing external input. Recycling mineral nutrients does not restore dissipated energy to its original high-quality form.

Why not the others: (A) mistakes use of stored chemical energy for creation of new energy. (B) turns nutrient cycling into complete energy recycling. (C) denies the energy available to consumers of dead organic material.

Key point: Distinguish branching food pathways and matter recycling from the one-way degradation of usable energy.

Chemistry

Q16 / A C2 molecular-orbital filling separates bond order from an assumed occupied 2p sigma bond

Answer: (A) The stated ordering gives a diamagnetic bond-order-two configuration without occupied sigma(2pz).

Given: C2 supplies eight valence electrons and the diagram states the applicable orbital ordering.

Formula/Principle: Bond order is half the excess of bonding over antibonding electrons; unpaired occupancy determines magnetism.

Step 1: Four electrons fill sigma(2s) and sigma*(2s), whose bond-order contributions cancel.

Step 2: The remaining four occupy the two equal-energy pi bonding orbitals as paired electrons. Sigma(2pz) remains empty.

Step 3:

$$\text{bond order} = (6-2)/2 = 2.$$

Why: This molecular-orbital description counts occupation of delocalised bonding and antibonding orbitals. It need not reproduce the simple localised rule that a drawn double bond is described as one sigma and one pi bond. Every occupied orbital here is paired.

Why not the others: (B) stops after single occupation instead of placing all four remaining electrons. (C) misidentifies the shown pi bonding levels as antibonding. (D) places electrons in a higher empty level without enough electrons.

Key point: Fill the stated MO ordering before importing a localised bond picture.

Q17 / B EDTA donor count and complex charge distinguished from colour and transition-metal status

Answer: (B) Six donor atoms give hexadentate coordination, and charge addition gives a dianion.

Why: Denticity counts donor atoms of the same ligand attached to one metal, not merely the number of negatively charged groups. The two nitrogen donors add to the four oxygen donors, so the stated binding mode is hexadentate. Combining Ca2+ with Y4− gives a complex charge of −2. Coordination is not restricted to transition metals, and visible colour is not a necessary consequence of every metal–ligand interaction.

Why not the others: (A) ignores four donors and fails to add charges. (C) excludes neutral nitrogen lone-pair donation and undercounts denticity. (D) makes a metal-class label cancel a real ionic charge.

Key point: Count attached donor atoms and add charges independently; colour is a separate electronic property.

Q18 / C Permanganate thermal decomposition couples oxygen evolution to two distinct manganese reductions

Answer: (C) One O2 molecule is formed per two KMnO4 units, with four electron equivalents transferred.

Given: The products are K2MnO4, MnO2 and O2.

Formula/Principle: Balance atoms and then equate oxidation-state increases and decreases.

Step 1: Balance potassium and manganese, then oxygen:

$$2\,\mathrm{KMnO_4} \longrightarrow \mathrm{K_2MnO_4} + \mathrm{MnO_2} + \mathrm{O_2}.$$

Step 2: The two manganese atoms change from +7 to +6 and +4, accepting

$$(7-6) + (7-4) = 4$$

electrons in total.

Step 3: Two oxygen atoms go from -2 to zero in O2, losing four electrons.

Why: Neutral formula units can contain elements undergoing oxidation-state changes. The internal electron balance links oxygen evolution to reduction of two manganese centres to different final states.

Why not the others: (A) doubles oxygen beyond atom balance. (B) supplies too little oxygen and too few electron equivalents. (D) confuses zero net formula charge with unchanged elemental oxidation states.

Key point: Verify both atom conservation and oxidation-state accounting in a thermal decomposition.

Q19 / D Period-3 configurations and basic–amphoteric–acidic oxide behaviour provide independent identification cues

Answer: (D) A basic sodium oxide, amphoteric aluminium oxide and acidic silicon oxide fit the observations.

Why: The three configurations identify sodium, aluminium and silicon in that order. Across the period the character of their typical oxides changes from strongly basic through amphoteric to acidic as metallic character decreases. Reaction with both acid and alkali is therefore the diagnostic contrast for aluminium oxide here. The claim concerns the usual acid–base behaviour of these oxides, not an assertion that all oxides in a block share one universal behaviour.

Why not the others: (A) assigns the alkali-reactive acidic oxide to the basic observation. (B) exchanges all three behaviours. (C) makes silicon oxide amphoteric and aluminium oxide purely acidic in these tests.

Key point: Combine valence configuration with the cross-period oxide trend rather than identifying an element from one reaction alone.

Q20 / B Unequal confinement and unequal mass give opposite rankings of velocity spread and kinetic-energy scale

Answer: (B) The ratios are 1000/1836 for velocity uncertainty and 1000000/1836 for the defined energy scale.

Given: Proton confinement is one-thousandth as wide, and its mass is 1836 times larger.

Formula/Principle:

$$\Delta x \Delta p \geq \hbar/2, \quad \Delta v = \Delta p/m, \quad K_* = (\Delta p)^2/(2m).$$

Step 1: At the stated lower bounds,

$$\Delta p_p/\Delta p_e = a/(a/1000) = 1000.$$

Step 2: Divide by the mass ratio:

$$\Delta v_p/\Delta v_e = 1000/1836.$$

Step 3: Square the momentum ratio before dividing by mass:

$$K_{*,p}/K_{*,e} = 1000^2/1836.$$

Why: A heavier particle can have a smaller velocity spread yet a larger kinetic-energy scale when it is confined much more tightly. These are lower-bound comparisons, not claims that every physical state saturates the uncertainty relation.

Why not the others: (A) ignores mass and momentum squaring. (C) reverses both requested ratios. (D) assumes equal momentum uncertainty despite unequal confinement.

Key point: Distinguish momentum, velocity and kinetic-energy scales when applying the uncertainty principle.

Q21 / A Retained water in a gravimetric product creates a directional sample-purity bias

Answer: (A) The dry precipitate gives 93.6% NaCl; including retained water would overestimate it.

Given: All chloride originates from NaCl, and each mole gives one mole of AgCl.

Formula/Principle: Convert the dry precipitate to moles and use the one-to-one chloride stoichiometry.

Step 1: Use only constant dry mass:

$$n(\text{AgCl}) = 1.148/143.5 = 0.00800 \text{ mol}.$$

Step 2:

$$m(\text{NaCl}) = 0.00800(58.5) = 0.468 \text{ g}.$$

Step 3:

$$\%\text{purity} = 100(0.468/0.500) = 93.6\%.$$

Why: Retained water adds measured mass without adding chloride ions. Dividing the wet mass by AgCl molar mass would therefore invent extra AgCl and extra original NaCl. The chloride-free impurity and complete recovery assumptions make the stoichiometric inference valid.

Why not the others: (B) reverses the direction of the wet-mass bias. (C) assumes perfect purity without using the chloride amount. (D) fails to convert precipitate mass into NaCl mass using their different molar masses.

Key point: A gravimetric product must have the assumed composition before its mass is converted to analyte amount.

Q22 / B Allylic-radical contributors preserve connectivity, charge and unpaired-electron number

Answer: (B) Resonance changes electron assignment while retaining the atomic skeleton and overall electron count.

Why: The three-carbon sigma framework is identical in both contributors. Moving the pi-electron and radical descriptions does not add or remove an electron, so the overall neutral charge and one unpaired electron are retained. The real allylic radical has delocalised electronic character rather than being a time average of independently isolable molecules. The drawings are bookkeeping contributors to one electronic structure.

Why not the others: (A) confuses electron redistribution with rotation of an atomic framework. (C) creates charges absent from the displayed electron counts. (D) treats resonance contributors as distinct chemical species in equilibrium.

Key point: Check connectivity, charge and electron count before interpreting a pair of resonance drawings.

Q23 / D Anti addition and internal symmetry determine different stereochemical sets from cis and trans alkenes

Answer: (D) Anti addition to the two alkene geometries gives different stereochemical sets.

Why: Each alkene carbon becomes a stereocentre after bromine addition. Adding from opposite faces to cis-but-2-ene produces the R,R and S,S pair, with equal access to the two mirror-related approaches in an achiral environment. Anti addition to trans-but-2-ene instead gives the R,S arrangement, which is identical to S,R by the molecule's internal symmetry and is meso. Thus a formal two-centre count must be filtered by both the addition constraint and molecular symmetry.

Why not the others: (A) reverses the anti-addition outcomes. (B) invents selection of a single handed product without a chiral influence. (C) ignores the meso identity and the stereospecific starting geometry.

Key point: Apply the reaction's relative stereochemistry before removing symmetry-equivalent product configurations.

Q24 / C Chloroform storage precautions distinguish oxidation prevention from phosgene trapping

Answer: (C) Prevention of oxidation and chemical trapping of its product are complementary precautions.

Why: Chloroform can be oxidised by air in the presence of light to carbonyl chloride, also called phosgene. Dark storage and restricted contact with air reduce the conditions favouring that change. The added ethanol reacts with phosgene, helping remove the hazardous product rather than merely lowering its concentration by a trivial volume addition. These roles explain why the precautions are combined rather than treating one as proof that no reactive impurity can ever form.

Why not the others: (A) reverses the role of darkness and invents simple reductive dechlorination. (B) assumes stoppering instantaneously eliminates pre-existing dissolved oxygen. (D) proposes complete salt formation that is not the stated storage chemistry.

Key point: Distinguish suppression of a side reaction from chemical removal of the side product.

Q25 / A Reimer–Tiemann formylation and Kolbe carboxylation lead to different diagnostic oxidation and acid-base responses

Answer: (A) Reimer–Tiemann formylation and Kolbe carboxylation introduce different one-carbon groups.

Step 1: Phenol with chloroform and alkali, then acid work-up, gives mainly salicylaldehyde, with CHO ortho to the hydroxyl group. Its introduced group is −CHO.

Step 2: Carbon dioxide reacts with sodium phenoxide under the specified conditions; acidification gives mainly salicylic acid, with −COOH.

Step 3: The aldehyde reduces Tollens reagent, whereas the carboxylic acid releases CO_2 from bicarbonate; phenolic acidity alone is insufficient for that bicarbonate test.

Why: Both routes add one carbon adjacent to the hydroxyl group, but reagent identity determines its oxidation state and functional group. Diagnostic tests must be applied to those actual groups rather than to carbon count or aromaticity alone.

Why not the others: (B) identifies different oxidation states as one product. (C) assigns groups formed by neither route. (D) wrongly exempts an aromatic aldehyde from the Tollens response and denies carboxylic acidity.

Key point: Track the introduced functional group before choosing a product test.

Q26 / B Protease sensitivity distinguishes peptide structure from steroid structure without changing the common signalling role

Answer: (B) Hormonal function does not imply one universal chemical composition.

Why: A peptide hormone contains a polypeptide structure whose integrity can matter for recognition and biological activity. Protease hydrolysis disrupts peptide bonds and can therefore abolish its activity. A steroid has a different carbon-ring framework rather than a peptide backbone, so the same enzyme need not destroy it. The matched treatment supports a chemical-class distinction while preserving the shared signalling role of the two hormones.

Why not the others: (A) makes all hormones proteins despite the explicitly different steroid structure. (C) infers inorganic composition from resistance to one selective enzyme. (D) confuses chemical messengers with bulk metabolic fuels and dismisses a relevant structural change.

Key point: A shared biological role does not require identical chemical bonds or susceptibility to an enzyme.

Q27 / C Competing proton donation and acceptance explain basic bicarbonate without a universal hydrogen-salt rule

Answer: (C) Bicarbonate can donate or accept a proton, with base behaviour stronger under this comparison.

Given: HCO3− has acidic constant Ka2 and conjugate-base constant Kw/Ka1.

Step 1:

$$K_b = 10^{-14}/(5 \times 10^{-7}) = 2 \times 10^{-8}.$$

Step 2: This exceeds

$$K_a = 5 \times 10^{-11}$$

by a factor of 400.

Why: Bicarbonate's charge does not forbid proton donation, but it can also accept a proton from water. Comparing those competing tendencies explains an alkaline rather than acidic salt solution in the stated regime. Na+ is a spectator ion for this hydrolysis.

Why not the others: (A) denies the retained acidic proton. (B) ignores the relative acid and base strengths. (D) assigns hydrolysis to the cation of a strong base.

Key point: An amphiprotic ion's name or charge alone does not determine its solution's pH.

Q28 / A Temperature-dependent Henry constant can cancel a simultaneous increase in gas partial pressure

Answer: (A) The dissolved mole fraction is unchanged, although several experimental variables change.

Given: Initial partial pressure is 0.20P; final total pressure is 2P and final gas fraction 0.40.

Formula/Principle: Dissolved mole fraction is partial pressure divided by the temperature-specific Henry constant.

Step 1:

$$p_2/p_1 = (0.40 \times 2P)/(0.20P) = 4.$$

Step 2:

$$x_2/x_1 = (p_2/p_1)(K_{H1}/K_{H2}) = 4/4 = 1.$$

Why: Henry's relation uses the partial pressure of the dissolving gas, not the entire gas mixture's pressure. Temperature also changes the proportionality constant here. The increased partial pressure is exactly offset by the increased KH, so neither total pressure nor temperature alone determines the outcome.

Why not the others: (B) ignores the composition change and overstates what total pressure establishes. (C) treats KH as unchanged despite the stated factor of four. (D) omits the KH change and denies partial-pressure dependence.

Key point: Separate total pressure, gas composition and the temperature-specific Henry constant.

Q29 / D Parallel electrolysis requires branch-specific charge rather than series-cell equality

Answer: (D) Copper deposition is 0.635 g and silver deposition 3.240 g.

Given: Parallel branches have different measured currents; equal time does not mean equal charge.

Formula/Principle:

$$n_e = It/F, \quad n_{\text{metal}} = n_e/z.$$

Step 1: For copper,

$$n_{e,Cu} = 0.20(9650)/96500 = 0.0200, \quad n_{Cu} = 0.0100 \text{ mol}.$$

Step 2: For silver,

$$n_{e,Ag} = 0.30(9650)/96500 = 0.0300, \quad n_{Ag} = 0.0300 \text{ mol}.$$

Step 3:

$$m_{Cu} = 0.0100(63.5) = 0.635 \text{ g}, \quad m_{Ag} = 0.0300(108) = 3.240 \text{ g}.$$

Why: Faraday accounting must use each branch's own charge and the correct number of electrons per deposited ion. The equality of charge in a series circuit does not apply to these parallel currents.

Why not the others: (A) omits copper's two-electron requirement. (B) wrongly gives silver a two-electron requirement. (C) divides total charge equally despite the unequal currents.

Key point: Identify electrical branch topology before applying electron-equivalent stoichiometry.

Q30 / D Stoichiometric coupling changes overall Gibbs feasibility without changing an isolated component's sign

Answer: (D) The combined Gibbs change is negative without changing R's uncoupled sign.

Given: One coupled event includes two copies of R and one of S under the same thermodynamic conditions.

Formula/Principle: Add Gibbs changes with the actual stoichiometric coefficients of the coupled process.

Step 1: Gibbs changes are extensive in reaction amount:

$$\Delta G_{coupled} = 2(20) + (-50) = -10 \text{ kJ}.$$

Step 2: The negative sum permits a favourable combined transformation if a mechanism actually couples the events.

Why: An energy-releasing component can drive an unfavourable component when their stoichiometric coupling is enforced. Merely placing separate reactants together does not prove that this mechanism exists, and it does not change the Gibbs change assigned to R in isolation.

Why not the others: (A) adds R only once and wrongly makes each part favourable. (B) reverses the sum and denies the role of coupling. (C) transfers the sign of the combined process to an uncoupled component.

Key point: Add reaction Gibbs changes with their actual coupling coefficients and retain the distinction between overall and component feasibility.

Mathematics

Q31 / C Concentric chord-length difference fixes line distance before an external point fixes two slopes

Answer: (C) The product is −16/9.

Given: If d is the common perpendicular distance from the origin, the half-chords are A=sqrt(25−d²) and B=sqrt(13−d²).

Formula/Principle: Chord length is twice the radius–distance right-triangle leg.

Step 1: A−B=2 and

$$A^2 - B^2 = 12 \Rightarrow A + B = 6.$$

Step 2: Thus A=4, B=2 and

$$d^2 = 25 - 16 = 9.$$

Step 3: A line through (0,5) has y=mx+5; the vertical line through that point has d=0 and fails. Hence

$$\frac{5}{\sqrt{1+m^2}} = 3 \Rightarrow m = \pm 4/3.$$

Worked derivation:

$$m_1 m_2 = -(4/3)^2 = -16/9.$$

Why: Both chords share one line-to-centre distance; their difference fixes that distance before the external-point condition fixes two orientations.

Why not the others: (A) inverts the squared slope. (B) loses the opposite signs. (D) reports one slope instead of their product.

Key point: Separate a shared metric constraint from the final orientation constraint.

Q32 / D Translated parabola reconstruction combines directrix, tangent slope and contact incidence

Answer: (D) The focus is (2,−1).

Given: Write the parabola as $(y-k)^2=4p(x-h)$, where $p>0$.

Formula/Principle: Its directrix is $x=h-p$ and its tangent slope satisfies $2(y-k)y'=4p$.

Step 1: The directrix gives

$$h - p = -1.$$

Step 2: At (2,2), $y'=1$, so

$$2 - k = 2p.$$

Step 3: Substitution of the point into the parabola gives

$$(2p)^2 = 4p(2 - h) \Rightarrow h = 2 - p.$$

Step 4: Combine the two h equations: $2-2p=-1$, so $p=3/2$, $h=1/2$ and $k=-1$.

Worked derivation:

$$F = (h + p, k) = (2, -1).$$

Why: Tangent incidence alone would not locate a translated parabola; its slope supplies a second local condition, and the directrix fixes the remaining translation/scale.

Why not the others: (A) is the vertex. (B) is the contact point, not the focus. (C) lies on the directrix and reverses the focus location.

Key point: Keep vertex translation and focal parameter separate when reconstructing a conic.

Q33 / A Logarithmic differentiation and second derivative establish the global minimum of a variable-base power

Answer: (A) Logarithmic differentiation gives one minimum and positive curvature.

Given: $x>0$ permits taking $\ln f = x \ln x$.

Formula/Principle: Differentiate the logarithm, then recover the derivatives of f.

Step 1:

$$f' = x^x(1 + \ln x).$$

Step 2: Since $x^x>0$, f' changes from negative to positive at

$$1 + \ln x = 0 \Rightarrow x = e^{-1}.$$

Step 3:

$$f'' = x^x \left[(1 + \ln x)^2 + 1/x\right] > 0.$$

Worked derivation:

$$f(e^{-1}) = (e^{-1})^{e^{-1}} = e^{-1/e}.$$

Why: The derivative's sign gives decrease before and increase after the point across the entire connected domain, establishing a global minimum. Positivity of the original function alone says nothing about monotonicity.

Why not the others: (B) reverses the derivative sign change and curvature. (C) includes an excluded endpoint and a nonstationary point. (D) confuses positive height with positive slope.

Key point: Use logarithmic differentiation when both base and exponent vary.

Q34 / D Pointwise ordering proves uniqueness of a parameter in a definite integral

Answer: (D) Exactly one value works, a=1.

Given: The denominator is positive for all x in [0,1] and a≥0.

Formula/Principle: Compare integrands pointwise before attempting to solve a transcendental equation.

Step 1: If b>a, then for every x>0,

$$\frac{1}{1+bx} < \frac{1}{1+ax},$$

so F(b)<F(a).

Step 2: At a=1,

$$F(1) = [\ln(1+x)]_0^1 = \ln 2.$$

Worked derivation: Strict decrease and this exhibited value imply

$$F(a) = \ln 2 \iff a = 1.$$

Why: Strict monotonicity prevents repeated parameter values even though a formula involving ln(1+a)/a would lead to a nonlinear equation. The single equality at x=0 does not remove strictness of the integral comparison.

Why not the others: (A) ignores the explicit working value. (B) contradicts strict decrease. (C) treats an integral identity at one value as an identity for every parameter.

Key point: A pointwise ordering can prove uniqueness without explicitly inverting an integral formula.

Q35 / A A rational three-cycle requires domain exclusions and a separate absence-of-fixed-points check

Answer: (A) The invariant real domain excludes 0 and 1, and every remaining point has exact period three.

Given: Each denominator in successive composition must remain nonzero.

Formula/Principle: Compose the rational map, then test smaller divisors of a returning iterate for fixed points.

Step 1: x=0 is excluded initially, and x=1 maps to zero and fails at the next step.

Step 2: Direct composition gives

$$f^2(x) = -\frac{1}{x-1}, \quad f^3(x) = x.$$

Step 3: A fixed point would satisfy

$$x = (x-1)/x \Rightarrow x^2 - x + 1 = 0,$$

whose discriminant is -3. There are no real fixed points.

Worked derivation: Since the period divides 3 and cannot be 1,

$$\text{least period} = 3.$$

Why: On the remaining domain no iterate can become zero or one, and the displayed cycle closes. A returning third iterate alone would not exclude fixed points; the separate quadratic check is necessary.

Why not the others: (B) confuses a three-cycle with an involution. (C) misses a later denominator and asserts fixedness. (D) uses a starting value whose next composition is undefined.

Key point: Check the invariant composition domain and the least period separately.

Q36 / B A column update is checked through determinant preservation and one inverse-column system

Answer: (B) The first inverse column is (−1/2,1/2,−1/2) transpose.

Given: The stated column change produces rows (1,3,0),(0,1,1),(1,2,1).

Formula/Principle: The first inverse column v solves Bv=e1; a column-addition operation preserves determinant.

Step 1: det B=det A=2, so the inverse exists.

Step 2: The equations are

$$v_1 + 3v_2 = 1, \quad v_2 + v_3 = 0, \quad v_1 + 2v_2 + v_3 = 0.$$

Step 3: The last two imply v3=−v2 and v1=−v2. The first then gives v2=1/2.

Worked derivation:

$$B(-1/2, 1/2, -1/2)^T = (1, 0, 0)^T.$$

Why: An inverse column is determined by its unit-vector image, so solving this one system is sufficient. Copying the old inverse or treating a column change as a row change would solve a different system.

Why not the others: (A) gives the opposite unit image. (C) is the original A inverse column before the change. (D) has twice the required scale.

Key point: Reconstruct the modified matrix, then solve the precise unit-image equation.

Q37 / D A reported minimum conditions ordered dice triples without selecting a labelled die

Answer: (D) The conditional probability is 6/61.

Given: Outcomes are ordered triples and all 216 original triples are equally likely.

Formula/Principle: Count the reported-minimum event before counting its favourable subset.

Step 1: Minimum exactly 2 means all results lie in {2,3,4,5,6} but not all in {3,4,5,6}:

$$N_E = 5^3 - 4^3 = 61.$$

Step 2: Subtract 2 from each result. For sum 8 the residual sum is 2, giving permutations of (2,0,0) or (1,1,0), three each.

Worked derivation:

$$P(\text{sum } 8 \mid E) = \frac{3+3}{61} = \frac{6}{61}.$$

Why: A minimum report does not identify which labelled die equals 2, and outcomes with multiple twos must not be overcounted. Counting the complementary all-above-two triples avoids that problem.

Why not the others: (A) treats unordered sum classes as equally likely. (B) retains triples whose minimum is above 2. (C) omits one of the two favourable partition types.

Key point: Translate an unlabelled summary observation into a complete event on the original ordered sample space.

Q38 / B Successive absolute-term ratios locate a tied maximum before alternating signs are restored

Answer: (B) The greatest magnitudes occur at r=2 and r=3, and their signed sum is zero.

Given: At x=1/4, term r is C(8,r)(−1/2)^r.

Formula/Principle: Compare successive absolute values before restoring signs.

Step 1:

$$\frac{|T_{r+1}|}{|T_r|} = \frac{8-r}{2(r+1)}.$$

Step 2: The ratio exceeds 1 for r<2, equals 1 at r=2 and is below 1 afterwards. Thus the tied maxima are r=2,3.

Step 3:

$$T_2 = 28/4 = 7, \quad T_3 = -56/8 = -7.$$

Worked derivation:

$$T_2 + T_3 = 7 - 7 = 0.$$

Why: The question compares evaluated term magnitudes, not raw binomial coefficients. Their equality in magnitude does not imply equality in sign because the substituted expansion alternates.

Why not the others: (A) chooses the wrong adjacent maximum and ignores signs. (C) adds magnitudes instead of signed terms. (D) stops before the ratio reaches unity.

Key point: Separate coefficient size, substituted term magnitude and algebraic sign.

Q39 / C Alternating AP sums require a parity check before reconstructing the last term

Answer: (C) The final term is 19.

Given: an=2n+1 and consecutive paired differences are −2.

Formula/Principle: Split the finite alternating sum by parity of n.

Step 1: If n=2k, the sum is −2k=−n and cannot equal positive 11.

Step 2: If n=2k+1, the sum is

$$-2k + (4k+3) = 2k + 3 = n + 2.$$

Step 3: n+2=11 gives n=9.

Worked derivation:

$$a_9 = 3 + 8(2) = 19.$$

Why: Pairing is different depending on whether an unpaired positive final term remains. A single formula obtained by pairing every term would miss the admissible odd case.

Why not the others: (A) reports the alternating sum itself. (B) uses one fewer progression term. (D) uses one extra term.

Key point: Resolve parity before applying a paired-sum pattern.

Q40 / A A removable algebraic cancellation changes at an exceptional parameter, so two limit operations differ

Answer: (A) The values are 0 and −1/2.

Given: For each fixed a the square root is defined in a neighbourhood of x=0.

Formula/Principle: Treat the exceptional parameter before cancelling factors and taking a later limit.

Step 1: At a=0, the numerator is exactly zero for x≠0, so L(0)=0.

Step 2: For a≠0, rationalisation gives

$$\frac{\sqrt{1+ax}-1}{x(x-a)} = \frac{a}{(x-a)(\sqrt{1+ax}+1)}.$$

Step 3: Now x→0 gives

$$L(a) = a/[(-a)2] = -1/2.$$

Worked derivation:

$$\lim_{a\to 0} L(a) = -1/2 \neq L(0).$$

Why: Fixing a nonzero parameter permits a cancellation and limit whose eventual expression is discontinuous at the exceptional parameter. Substituting a=0 into an expression derived only for a≠0 loses that distinction.

Why not the others: (B) extends the nonzero-parameter result to the excluded case. (C) extends the exceptional zero numerator to nearby nonzero parameters. (D) loses the negative sign and exchanges the limits.

Key point: Parameter limits need not commute with the limiting operation defining a family.

Q41 / C Parameter rank loss preserves a common plane-intersection line before its nearest point is found

Answer: (C) A common line occurs for a=2,b=1 and has distance sqrt(6)/4.

Given: The first two planes are independent.

Formula/Principle: An independent third-plane equation selects a point; an identically satisfied one preserves a line.

Step 1: Subtract their equations to obtain y=1/2, then x+z=1/2.

Step 2: On this line, the third equation becomes

$$1 + (a-2)z = b.$$

It holds identically exactly when a=2,b=1. Otherwise it selects one point if a≠2 or no point if a=2,b≠1.

Step 3: On the line, minimise

$$d^2 = x^2 + z^2 + 1/4$$

subject to x+z=1/2. The minimum is at x=z=1/4.

Worked derivation:

$$d = \sqrt{1/16 + 1/16 + 1/4} = \sqrt{6}/4.$$

Why: An identically satisfied third equation preserves the first two planes' entire line; a merely consistent equation generally leaves only one point.

Why not the others: (A) omits the required constant consistency. (B) gives an independent third-plane constraint. (D) uses a nonminimal point on the correct line.

Key point: Determine the dimension of intersection before computing a geometric distance.

Q42 / B Opposite bead positions and reflection identification require two separate circular-symmetry reductions

Answer: (B) There are 12 necklaces.

Given: Beads are all distinct, and both rotational and mirror copies are identified.

Formula/Principle: Remove rotations first, then check whether reflection pairs can have a fixed arrangement.

Step 1: Fix A at one vertex. B must occupy the opposite vertex, leaving

$$4! = 24$$

arrangements of the other beads.

Step 2: Reflection through the A–B axis exchanges the other four vertices in two pairs. Since their beads are distinct, no arrangement equals its reflection.

Worked derivation:

$$N = 24/2 = 12.$$

Why: Anchoring a labelled bead removes the six possible rotations exactly once. A further division by two is valid because no allowed labelled arrangement has mirror symmetry; that condition must be checked rather than assumed for an arbitrary repeated-bead necklace.

Why not the others: (A) retains reflections as distinct. (D) double-counts an orientation already fixed. (C) divides by reflection twice.

Key point: Quotient geometric arrangements only after checking whether symmetries fix any labelled configuration.

Q43 / A A conjugate equation factors into real-axis and unit-circle locus components

Answer: (A) Either z is real and nonzero, or its modulus is one.

Given: Reality means equality to the complex conjugate, and z≠0.

Formula/Principle: A real complex expression equals its conjugate; retain every factor when solving the resulting equation.

Step 1:

$$z + 1/z = \bar{z} + 1/\bar{z}.$$

Step 2: Rearrange without discarding either factor:

$$(z - \bar{z})\left(1 - \frac{1}{|z|^2}\right) = 0.$$

Step 3: The first factor gives real z; the second gives |z|=1. Both sets satisfy the original condition, with zero still excluded.

Worked derivation: For z=re^(i theta),

$$\mathrm{Im}(z + 1/z) = (r - r^{-1})\sin\theta = 0.$$

Why: Dividing by either factor too early would remove an entire component. The real-axis and circular branches overlap at ±1 but their union is the complete locus.

Why not the others: (B) discards nonunit real solutions. (C) includes imaginary values whose reciprocal sum is generally nonreal. (D) includes an undefined zero and misses nonreal unit-modulus solutions.

Key point: A factorised complex constraint can define a union rather than one geometric curve.

Q44 / B A zero branch can join a translated positive branch smoothly, invalidating naive initial-value uniqueness

Answer: (B) Delayed departure from the zero solution gives infinitely many valid solutions.

Given: The differential equation has first order and degree two; differentiability, not a preselected sign of y', is required.

Formula/Principle: Check each proposed branch and the join directly in the original differential equation before dividing by a vanishing factor.

Step 1: On a positive increasing branch, y'=2sqrt(y) integrates to

$$\sqrt{y} = x - c, \quad y = (x - c)^2 \quad (x \geq c).$$

Step 2: On the earlier interval choose y=0, whose derivative is zero and satisfies the original equation.

Step 3: At x=c, both values and both one-sided derivatives are zero, so the joined function is continuously differentiable.

Worked derivation: On the positive branch,

$$(y')^2 = [2(x - c)]^2 = 4(x - c)^2 = 4y;$$

on the zero branch both sides vanish.

Why: Separation by sqrt(y) can miss solutions that remain at y=0 for an interval. Direct substitution and checking the join establish nonuniqueness without relying on an inapplicable uniqueness assumption.

Why not the others: (A) ignores every delayed departure. (C) assumes an initial zero derivative forces perpetual rest. (D) mistakes failure of one division step for failure of the equation.

Key point: Check singular branches and joins in the original differential equation.

Q45 / C Different principal inverse ranges give an interval of equality plus one isolated endpoint

Answer: (C) The equality holds on [0,pi/2] and at the isolated endpoint 2pi.

Given: The two principal inverse ranges are different.

Formula/Principle: Evaluate each inverse composition in its stated principal range.

Step 1: On [0,pi/2], both inverse compositions return theta.

Step 2: On (pi/2,pi], they return pi−theta and theta respectively, which cannot be equal except at the already included pi/2.

Step 3: On (pi,2pi), arcsin(sin theta) is negative while arccos(cos theta) is positive. At 2pi both become zero.

Worked derivation:

$$\{\theta : \text{equality}\} = [0, \pi/2] \cup \{2\pi\}.$$

Why: The same angle input is folded into different principal ranges, so cancelling an inverse against its forward function globally is invalid. Signs resolve the entire open lower-half-circle interval without extra algebra.

Why not the others: (A) ignores the fold after pi/2. (B) omits a whole interval and an endpoint. (D) compares negative and positive principal values.

Key point: Resolve inverse-function branches before comparing compositions.

Physics

Q46 / D Constant-force and constant-power braking produce different initial-speed distance scalings

Answer: (D) The constant-force distance scales by 4 and the constant-power distance by 8.

Given: Both F and P are held fixed when the initial speed is changed.

Formula/Principle: Constant-force work equals kinetic-energy loss; constant removed power is the time rate of that loss.

Step 1: Constant-force work gives

$$F s_F = \tfrac{1}{2} m v_0^2 \Rightarrow s_F \propto v_0^2.$$

Step 2: For constant removed power,

$$mv\,dv/dt = -P, \quad ds = v\,dt = -mv^2\,dv/P.$$

Step 3: Integrating to rest gives

$$s_P = \frac{m}{P}\int_0^{v_0} v^2 dv = \frac{m v_0^3}{3P}.$$

Worked derivation:

$$s_F(2v_0)/s_F(v_0) = 4, \quad s_P(2v_0)/s_P(v_0) = 8.$$

Why: Constant power requires a speed-dependent braking force P/v rather than constant deceleration. Its ideal force diverges near rest, so the result belongs to the stipulated mathematical model.

Why not the others: (A) assumes distance scales linearly with speed. (B) applies constant-force scaling to both. (C) exchanges the two models.

Key point: Identify what is constant before choosing a stopping-distance law.

Q47 / C Dimensionless exponential arguments and additive power terms constrain coefficients without proving a proposed law

Answer: (C) The required dimensions are M T−1, L−1 and M L T−2, but the law is not established by that check.

Given: Power has dimensions M L² T−3; speed is L T−1, acceleration L T−2 and displacement L.

Formula/Principle: An exponential argument is dimensionless, and every term added as a power must have power dimensions.

Step 1: From beta x dimensionless,

$$[\beta] = L^{-1}.$$

Step 2: The first term requires

$$[\alpha] = [P]/[v]^2 = (ML^2T^{-3})/(L^2T^{-2}) = MT^{-1}.$$

Step 3: Since sqrt(a x) has speed dimensions,

$$[\gamma] = [P]/[\sqrt{ax}] = (ML^2T^{-3})/(LT^{-1}) = MLT^{-2}.$$

Worked derivation: Each term has dimensions

$$(MT^{-1})(L^2T^{-2}) = (MLT^{-2})(LT^{-1}) = ML^2T^{-3}.$$

Why: Dimensionless arguments are required for transcendental functions, and summed terms must describe the same physical quantity. These restrictions can reject an invalid proposal but cannot fix all dimensionless factors or demonstrate that its assumed dependence matches experiments.

Why not the others: (A) makes beta x dimensional and overclaims physical proof. (B) assigns the entire power dimension to alpha without removing v² and gives gamma the wrong dimension. (D) miscounts time powers and treats a necessary test as sufficient proof.

Key point: Check transcendental arguments and every additive term, then distinguish consistency from verification.

Q48 / A Drop coalescence links volume conservation to different Stokes-speed and surface-energy scalings

Answer: (A) Terminal speed increases fourfold, while total surface energy halves.

Given: Volume is conserved and surface energy is gamma times area for a liquid–fluid interface.

Formula/Principle: Conserve volume; Stokes terminal speed scales as radius squared and surface energy as total area.

Step 1:

$$R^3 = 8r^3 \Rightarrow R = 2r.$$

Step 2: Stokes terminal speed is proportional to radius squared at fixed material parameters:

$$v_f/v_i = R^2/r^2 = 4.$$

Step 3: Initial total area is 8(4pi r²); final area is 4pi(2r)².

Worked derivation:

$$U_f/U_i = \frac{4\pi(2r)^2}{8(4\pi r^2)} = 1/2.$$

Why: A larger radius changes drag–weight balance and surface-to-volume ratio differently. Coalescence does not preserve total area, and the released surface energy need not remain as steady translational kinetic energy.

Why not the others: (B) uses linear-radius speed scaling. (C) reverses the surface-energy change. (D) equates combined volume factor to speed factor.

Key point: Apply volume conservation before using distinct radius scalings for motion and surface energy.

Q49 / B A sticking corner impact uses pivot angular momentum before a separate gravitational barrier check

Answer: (B) The threshold speed is sqrt(3gR/2).

Given: The corner impulse has zero moment about the corner, but the impact is not energy conserving.

Step 1: Before impact, angular momentum magnitude about the corner is

$$L = mv(R - R/2) + (mR^2/2)(v/R) = mvR.$$

Step 2: The new pivot inertia is

$$I_P = mR^2/2 + mR^2 = 3mR^2/2, \quad \omega' = L/I_P = 2v/(3R).$$

Step 3: Remaining kinetic energy is

$$K' = \tfrac{1}{2} I_P \omega'^2 = mv^2/3.$$

The centre must gain height R/2.

Worked derivation:

$$mv^2/3 \geq mgR/2 \Rightarrow v_{min} = \sqrt{3gR/2}.$$

Why: Angular momentum is conserved through the impulsive contact, then mechanical energy is conserved only during the subsequent ideal pivoting stage. The minimum is the limiting just-reaching condition.

Why not the others: (A) misses impact energy loss. (C) uses the incoming rolling energy as fully available. (D) keeps only incoming rotational energy and skips the impact angular-momentum balance.

Key point: Use the conserved quantity appropriate to each stage of an impact-and-climb process.

Q50 / A Unregenerated isochoric heating changes the efficiency of a two-isotherm cycle

Answer: (A) Efficiency is ln2/(3/2+2 ln2).

Given: Cv=3R/2; the two isochoric legs do no work.

Formula/Principle: Cycle efficiency is net work divided by all positive external heat input.

Step 1: The isothermal work contributions are

$$W_h = 2RT \ln 2, \quad W_c = -RT \ln 2, \quad W_{net} = RT \ln 2.$$

Step 2: Positive heat enters during isochoric heating and hot expansion:

$$Q_{in} = \tfrac{3}{2} RT + 2RT \ln 2.$$

Step 3: Both cooling legs reject heat and are not subtracted from the input definition.

Worked derivation:

$$\eta = \frac{W_{net}}{Q_{in}} = \frac{\ln 2}{3/2 + 2 \ln 2} \approx 0.240.$$

Why: Without regeneration, the isochoric heating cost must be supplied externally each cycle. Ignoring that cost would assign a different engine with internal heat recovery. Quasistatic does not by itself make the efficiency the Carnot value.

Why not the others: (B) omits the unregenerated isochoric heat cost. (C) miscounts the monatomic heat capacity and hot isothermal factor. (D) ignores unavoidable rejected heat.

Key point: Specify which heat is recovered internally before calculating cyclic efficiency.

Q51 / C Partial immersion adds buoyant stiffness and imposes an amplitude-dependent boundary on SHM

Answer: (C) Buoyancy adds restoring stiffness, and the proposed amplitude crosses an immersion boundary.

Given: Upward displacement u decreases submerged depth from h0 to h0−u while the block remains partly immersed.

Step 1: Relative to equilibrium, spring force decreases upward by ku and buoyancy decreases upward by rho g A u. Thus

$$F_{net} = -(k + \rho g A)u.$$

Step 2:

$$\rho g A = 1000(10)(0.010) = 100 \ \mathrm{N\,m^{-1}}, \quad \omega = \sqrt{200/2} = 10 \ \mathrm{rad\,s^{-1}}.$$

Step 3: A symmetric amplitude must satisfy Aosc<h0 and Aosc<H−h0 to remain strictly partly immersed. Here the tighter bound is 0.050 m.

Worked derivation:

$$u = 0.060 \ \mathrm{m} \Rightarrow h = h_0 - u = -0.010 \ \mathrm{m},$$

which is outside the model.

Why: Force balance removes constant terms but not the displacement dependence of buoyancy. Once fully emerged or fully submerged, its slope changes and the single linear law fails.

Why not the others: (A) ignores buoyant stiffness and the boundary. (B) confuses equilibrium with zero restoring slope. (D) leaves out a displacement-dependent force.

Key point: Linearise all position-dependent forces and check that the full amplitude remains within that regime.

Q52 / B Successive air-column spacing determines wavelength before an end correction uses the first mode

Answer: (B) The sound speed is 340 m s−1 and the correction is 0.050 m.

Given: Successive closed–open resonances differ in effective length by lambda/2.

Formula/Principle: The first two closed–open resonances have effective lengths lambda/4 and 3lambda/4.

Step 1:

$$\lambda/2 = 0.70 - 0.20 = 0.50 \ \mathrm{m} \Rightarrow \lambda = 1.00 \ \mathrm{m}.$$

Step 2:

$$v = f\lambda = 340(1.00) = 340 \ \mathrm{m\,s^{-1}}.$$

Step 3: For the first resonance, L1+e=lambda/4; taking the lower resonance as the fundamental gives

$$e = 0.25 - 0.20 = 0.050 \ \mathrm{m}.$$

Worked derivation: The next predicted length is

$$3\lambda/4 - e = 0.75 - 0.05 = 0.70 \ \mathrm{m}.$$

Why: The common end correction cancels in the spacing but must be restored to infer an absolute resonance length. The stated low-length pair is understood as the first two resonances; otherwise spacing alone fixes speed but not the harmonic index.

Why not the others: (A) uses quarter-wave spacing. (C) fits spacing but not the absolute first length. (D) uses full-wave spacing and an inconsistent correction.

Key point: Separate the correction-free spacing measurement from the mode-index-dependent absolute length.

Q53 / C Equatorial dipole potential cancels while a nonzero vector field remains

Answer: (C) Opposite charge potentials cancel, but their field vectors do not cancel.

Given: The point is equidistant from the two opposite charges and lies far from their separation scale.

Formula/Principle: Add scalar charge potentials and vector field components separately.

Step 1: Scalar potentials add to zero on the equatorial plane.

Step 2: The transverse field components cancel while the components opposite the dipole moment add.

Worked derivation:

$$V = \frac{kp\cos(\pi/2)}{r^2} = 0, \quad \mathbf{E} = -\frac{k\mathbf{p}}{r^3} \neq 0.$$

Why: A zero potential value does not require a zero spatial gradient. Moving away from the equatorial plane changes the relative charge contributions even though the value on that plane is zero.

Why not the others: (A) treats cancellation of scalars as cancellation of vectors. (B) reverses the actual cancellations. (D) assigns a direction to scalar potential and reverses the equatorial field.

Key point: Evaluate scalar potential and vector field by their respective addition rules.

Q54 / B Temperature-dependent resistance gives opposite thermal feedback under voltage and current control

Answer: (B) Voltage control yields a finite rise, whereas the ideal current model's feedback prevents a positive steady balance.

Given: R=10+0.10 delta T and heat balance requires electrical power equal to 0.10 delta T.

Formula/Principle: At steady temperature, electrical heating equals heat loss; use V^2/R or I^2R for the actual control mode.

Step 1: At constant voltage,

$$0.10\Delta T = 144/(10 + 0.10\Delta T).$$

Step 2: Hence

$$\Delta T^2 + 100\Delta T - 14400 = 0 \Rightarrow \Delta T = 80 \text{ K}$$

after rejecting the negative root.

Step 3: At constant current,

$$0.10\Delta T = 1.44(10 + 0.10\Delta T) \Rightarrow -0.044\Delta T = 14.4,$$

which has no nonnegative solution.

Worked derivation: At the voltage solution,

$$R = 18\ \Omega, \quad P = 144/18 = 8\ \text{W} = 0.10(80).$$

Why: Increasing resistance reduces V^2/R heating but increases I^2R heating. This model predicts failure of a steady balance under the current setting, not a temperature extrapolation beyond the model's validity.

Why not the others: (A) ignores different feedback after warming. (C) exchanges the power laws. (D) ignores the initial positive heating.

Key point: Electrical control mode changes thermal feedback even at equal initial power.

Q55 / D Initial stable daughter abundance changes the parent survival inferred from a later isotope ratio

Answer: (D) One-quarter of the initial parents remains, corresponding to two half-lives.

Given: The initial daughter population is not zero, and NP+ND is conserved.

Formula/Principle: Parent plus stable daughter number is conserved, including the initial daughters.

Step 1: Write the initial parent count as N0. Initial total nuclei are

$$N_0 + 2N_0 = 3N_0.$$

Step 2: At the later ratio 11, total nuclei are NP+11NP=12NP. Thus

$$N_P/N_0 = 3/12 = 1/4.$$

Step 3: Exponential decay gives

$$(1/2)^{t/T} = 1/4 \Rightarrow t/T = 2.$$

Worked derivation: Later daughters are

$$2N_0 + (N_0 - N_0/4) = 11N_0/4,$$

which reproduces the ratio 11.

Why: The daughter count includes inherited daughter nuclei and newly produced ones. Treating the final ratio as solely accumulated decay products would overestimate elapsed decay.

Why not the others: (B) ignores the original daughter contribution. (C) divides the two ratios without conserving total nuclei. (A) identifies the daughter ratio with the reciprocal surviving fraction.

Key point: Include stable initial products in decay conservation before extracting a clock time.

Q56 / D A translating rod in a transverse field gradient requires matching emf and force integrals

Answer: (D) Emf is 0.30 V, the lower end is positive, and the required external force is 0.10 N rightward.

Given: The nonuniform field must be integrated along the moving rod; v cross B points toward −y.

Step 1:

$$\mathcal{E} = v\int_0^L B_0(y/L)^2 dy = vB_0L/3 = 0.30\ \text{V}.$$

Step 2: Positive charges are driven toward y=0; current magnitude is

$$I = \mathcal{E}/R = 1.0\ \text{A}.$$

Step 3: Magnetic force opposes motion, with magnitude

$$F_B = I\int_0^L B(y)dy = IB_0L/3 = 0.10\ \text{N}.$$

Worked derivation:

$$F_{ext}v = 0.10(3) = 0.30\ \text{W} = \mathcal{E}^2/R.$$

Why: Replacing the field by its endpoint value triples both the emf and the relevant field integral. The power check confirms the resisting force direction and energy conversion to heat.

Why not the others: (A) uses endpoint field and reverses polarity. (B) reverses both polarity and external-force direction. (C) uses endpoint field for emf and inconsistent force accounting.

Key point: Integrate the actual motional field and use the same integral in the magnetic-force balance.

Q57 / B Immersion changes lens power through relative index and can reverse a convex lens's sign

Answer: (B) The focal lengths are +80 cm and −160 cm.

Given: The curvature factor is 1/20−1/(−20)=0.10 cm−1.

Formula/Principle:

$$1/f = (n_g/n_m - 1)(1/R_1 - 1/R_2).$$

Step 1: In index 4/3, ng/nm=9/8, so

$$1/f_1 = (1/8)(0.10) = 1/80 \text{ cm}^{-1}.$$

Step 2: In index 1.60, ng/nm=15/16, so

$$1/f_2 = (-1/16)(0.10) = -1/160 \text{ cm}^{-1}.$$

Worked derivation:

$$f_1 = 80 \text{ cm}, \quad f_2 = -160 \text{ cm}.$$

Why: Convex shape alone does not guarantee convergence; refraction depends on the lens index relative to its surrounding medium. The higher-index liquid reverses the sign of the power.

Why not the others: (A) uses the air formula twice. (C) reverses both power signs. (D) mishandles the two-surface curvature factor.

Key point: Replace absolute index by relative index when the optical environment changes.

Q58 / D Parallel RC currents add in quadrature while only the resistor consumes mean power

Answer: (D) The supply current is sqrt(2) A and the real power is 10 W.

Given: Parallel branch voltages are equal; resistor current is in phase with voltage and capacitor current leads by pi/2.

Formula/Principle: Parallel branch currents add as phasors, and real power is the cycle-averaged energy dissipation.

Step 1: Add the perpendicular current phasors:

$$I_{rms} = \sqrt{1^2 + 1^2} = \sqrt{2} \text{ A}.$$

Step 2: Only the ideal resistor consumes mean power:

$$P = VI_R = 10(1) = 10 \text{ W}.$$

Worked derivation: The total power factor is

$$\cos\phi = 1/\sqrt{2}, \quad VI\cos\phi = 10 \text{ W}.$$

Why: Current magnitudes are not in phase, so they cannot be added arithmetically. Capacitive energy storage returns energy during each cycle and does not contribute net dissipation in this ideal model.

Why not the others: (A) adds current magnitudes and apparent powers. (B) omits the capacitor contribution to total current. (C) ignores the resistor's dissipation.

Key point: In parallel AC circuits add current phasors, then distinguish apparent from real power.

Q59 / A Intrinsic carrier equality does not require equal electron and hole transport contributions

Answer: (A) Carrier numbers and carrier mobilities are different factors in conduction.

Given: Intrinsic generation creates equal electron and hole populations in Q, but their mobilities need not be equal.

Formula/Principle: Conductivity is the sum of electron and hole concentration–mobility contributions.

Step 1: P's band and temperature behaviour fit a metal; Q's small gap and paired carriers fit an intrinsic semiconductor; R's large gap fits an insulator in the stated range.

Step 2: The contributions satisfy

$$\sigma = e(n\mu_e + p\mu_h).$$

Worked derivation: With n=p,

$$\sigma_e/\sigma_h = \mu_e/\mu_h > 1.$$

Why: Intrinsic describes how carriers are generated and their concentrations, not equality of transport mobility. Temperature effects reflect competition between carrier generation and scattering, so the sign of the metal's temperature coefficient must be interpreted with its band evidence.

Why not the others: (B) infers donor doping from mobility alone. (C) conflates population equality with mobility equality. (D) reverses the supplied metal classification.

Key point: Use band structure, carrier source and mobility as separate pieces of evidence.

Q60 / C Medium wavelength and measured focal length can compensate in a diffraction-width comparison

Answer: (C) The two changes cancel and central width remains w.

Given: Diffraction depends on wavelength in the surrounding medium, and the new focal length is supplied rather than assumed unchanged.

Formula/Principle: Central diffraction width is twice focal length times wavelength in the medium divided by slit width.

Step 1: In air, the first zeros give

$$w = 2f\lambda_0/a.$$

Step 2: In the liquid,

$$\lambda_m = \lambda_0/1.50, \quad f_m = 1.50f.$$

Worked derivation:

$$w_m = \frac{2(1.50f)(\lambda_0/1.50)}{a} = w.$$

Why: The slit determines an angular scale while the lens maps that angle into a focal-plane distance. Immersion changes both pieces here, and using only one change would predict an incorrect width. The paraxial and far-field conditions are retained as stated.

Why not the others: (A) changes focal length only. (B) changes wavelength only. (D) multiplies by both factors instead of reducing wavelength.

Key point: Separate angular diffraction from the optical mapping onto a measurement plane.
